

**CSE218 Final Quiz****19 February 2022**

Full Marks: 40, Time: 40 mins (answer: 35 mins and scan and upload: 5 mins)

Instructions:

- i. There are eight (8) questions. Each question carries 5 marks.
- ii. Answer all the questions.
- iii. Answer each question in a separate page. Do not mix answers of multiple questions in the same page.

Answer all the questions below.

- ✓ 1. Use Gauss elimination with partial pivoting to solve the following system of linear equations.

$$4x_1 + x_2 - x_3 = -2$$

$$5x_1 + x_2 + 2x_3 = 4$$

$$6x_1 + x_2 + x_3 = 6$$

- ✓ 2. Find the root of the equation  $x^2 = 0.5 + \sin(x)$  in the interval  $[-1, 0]$ . Use the interval endpoints as the initial guesses. Show the first 5 iterations of the bisection method. For each iteration, show the estimate of the root and absolute approximate relative error.

- ✓ 3. Find the approximate value of  $\sqrt[3]{10}$  using Newton-Raphson method. Use 3 as the initial guess. Show the Newton-Raphson update equation for the given problem. Show the estimate of the root and absolute approximate relative error for the first three iterations.

- ✓ 4. Consider the given dataset below. Now -

- Find the value of  $f(x)$  using the **Newton's divided difference method for quadratic interpolation** where,

$$x = 12 + 3 * ((\text{last three digit of your student no} - 1) \% 30)$$

As an example, if your student no is 1905125, the value of  $x$  will be,

$$x = 12 + 3 * ((125 - 1) \% 30) = 24$$

- Find the **absolute approximate relative error** for the  $f(x)$  value computed above.
- Show detailed calculation steps including the divided difference table.

| x  | f(x)  |  | x  | f(x)  |
|----|-------|--|----|-------|
| 10 | 28.33 |  | 58 | 6.25  |
| 13 | 38.22 |  | 61 | 20.13 |
| 16 | 22.8  |  | 64 | 50.56 |

**CSE218 Final Quiz**  
**19 February 2022**

Full Marks: 40, Time: 40 mins (answer: 35 mins and scan and upload: 5 mins)

|    |       |  |     |       |
|----|-------|--|-----|-------|
| 19 | 30.36 |  | 67  | 24.66 |
| 22 | 37.98 |  | 70  | 6.36  |
| 25 | 57.25 |  | 73  | 74.19 |
| 28 | 41.87 |  | 76  | 82.23 |
| 31 | 55.67 |  | 79  | 13.8  |
| 34 | 33.92 |  | 82  | 70.48 |
| 37 | 92.81 |  | 85  | 66.58 |
| 40 | 80.46 |  | 88  | 16.65 |
| 43 | 29.46 |  | 91  | 77.41 |
| 46 | 32.08 |  | 94  | 41.41 |
| 49 | 72.48 |  | 97  | 88.66 |
| 52 | 62.51 |  | 100 | 62.43 |
| 55 | 79.68 |  | 103 | 40.51 |

5. Compare **Newton's divided difference** and **Lagrangian** methods of interpolation in terms of their respective advantages and disadvantages.

6. You need to find the  $n$ -th term of the following sequence:

14, 68, 158, ...

It is given that the term will be a polynomial expression. Which numerical method will you use to solve the problem? Show your work step by step.

7. Use Simpson's rule with  $n = 8$  to approximate the integral below

$$I = \int_2^6 f(x)$$

A table of values for the function  $f(x)$  is given below.

|        |      |      |      |      |       |       |       |       |       |       |
|--------|------|------|------|------|-------|-------|-------|-------|-------|-------|
| $x$    | 1.5  | 1.75 | 2    | 2.25 | 2.5   | 2.75  | 3     | 3.25  | 3.5   | 3.75  |
| $f(x)$ | 1.58 | 3.06 | 5.04 | 7.6  | 10.75 | 14.54 | 18.97 | 24.01 | 29.63 | 35.73 |

|        |       |       |       |       |       |       |      |       |       |       |
|--------|-------|-------|-------|-------|-------|-------|------|-------|-------|-------|
| $x$    | 4     | 4.25  | 4.5   | 4.75  | 5     | 5.25  | 5.5  | 5.75  | 6     | 6.25  |
| $f(x)$ | 42.16 | 48.72 | 55.12 | 60.94 | 65.63 | 68.48 | 68.5 | 64.43 | 54.63 | 36.94 |

**CSE218 Final Quiz**  
**19 February 2022**

Full Marks: 40, Time: 40 mins (answer: 35 mins and scan and upload: 5 mins)

- ✓ 8. You have to examine the relationship between the age and price for used cars sold in the last year by a car dealership company. Here is the table of the data:

| Car age (in years) | Price (in dollars) |
|--------------------|--------------------|
| 4                  | 5800               |
| 5                  | 5700               |
| 7                  | 4200               |
| 8                  | 4100               |
| 9                  | 3100               |
| 10                 | 2500               |

Which model fits the data best? Find the corresponding coefficients and hence the equation.